

Kaifeng “Kevin” Jia

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EDUCATION

Cornell University

Aug 2025 – Dec 2026 (Expected)

Master of Engineering in Electrical and Computer Engineering

- Selected coursework: Fast Robots, Autonomous Mobile Robots, Legged Robot, Deep Learning for Robotics.

Hefei University of Technology

Sep 2021 – Jun 2025

Bachelor of Engineering in Optoelectronic Information Science and Engineering

- Selected coursework: Signals and Systems, Digital Image Processing, Sensor Principles and Technology.

TECHNICAL SKILLS

Machine Learning & RL: Deep learning, reinforcement learning, multi-agent RL, PPO

Programming: Python, C/C++, Swift

Robotics & Perception: SLAM, EKF, Bayes filtering, PID control, sensor fusion (IMU/ToF), planning, YOLO

Simulation & Tools: Isaac Lab / Isaac Sim, PyTorch, Git, Linux, MATLAB

RESEARCH EXPERIENCE

Decentralized Multi-Agent Learning for UAV Ramp Admission

Aug 2026 – Present

Gao Labs, Cornell University

Advised by Yishuo Jiang

- Investigate decentralized ramp admission and coordination for heterogeneous UAVs in air corridors, examining how information-sharing constraints affect safety and task completion.
- Designed extensions to graph-based multi-agent reinforcement learning, incorporating heterogeneous maneuvering limits and information-availability masks for autonomous entry and merging.

Cooperative Humanoid Loco-Manipulation from Human Demonstrations

Jun 2026 – Present

Carnegie Mellon University

Advised by Hao E. Zhang

- Develop a framework for learning cooperative humanoid loco-manipulation from human demonstrations through OmniRetarget-based retargeting, embodiment adaptation, and reference-guided multi-agent reinforcement learning.
- Developed OmniRetarget-based methods for OMOMO and CORE4D, including two-stage object-scale adaptation, palm-orientation optimization, and synchronized dual-robot motion references.

Depth Foundation Model Adaptation via Nanophotonic Wavefront

May 2026 – Aug 2026

Prompting

New York University

Advised by Bingxuan Li (PhD Student)

- Explore architectures and adaptation modules for single-shot metric depth estimation from metalens polarization signals, building on nanophotonic wavefront prompting.
- Implemented polarization-data loading and preprocessing to convert multi-channel optical measurements into model-compatible inputs.
- Developed two optical-signal encoding modules to map polarization measurements into representations compatible with a pretrained depth backbone.
- Develop a training and evaluation framework covering model training, validation, and inference.

Indoor AR Navigation System for Olin Library

Sep 2025 – May 2026

Cornell University

- Built an iOS AR indoor navigation system using ARKit visual-inertial SLAM to estimate device pose and render real-time navigation guidance on the camera feed.
- Implemented relocalization using prerecorded visual feature maps, enabling automatic pose recovery upon re-entry without manual initialization.
- Constructed a metric-scale walkable graph from floor plans, applied Dijkstra's algorithm for path planning, and implemented angular-threshold turn detection with 3D AR directional cues.

Fast Robots: Autonomous RC Car Platform

Jan 2026 – May 2026

Cornell University

- Built an autonomy stack with ToF-based 360° mapping, Bayes-filter localization on a real robot, and navigation through nine waypoints using orientation PID and an onboard state machine.
- Designed distance and orientation PID controllers with Kalman-filter state estimation using identified drag and momentum parameters; implemented an autonomous ToF-triggered drift maneuver.
- Integrated VL53L1X ToF sensors, an ICM-20948 IMU, motor drivers, and BLE on an Artemis microcontroller for real-time sensor streaming and command execution.

Smart Desk Assistant: Multimodal Raspberry Pi System

Aug 2025 – Dec 2025

Cornell University

- Built a Raspberry Pi assistant integrating touch and voice interfaces, LLM-based dialogue, and PID-controlled face tracking, with a state machine managing wake-word detection and speech interaction.

Smart Prosthetic Sensing with Triboelectric Sensors

Apr 2023 – May 2024

Hefei University of Technology

Advised by Prof. Jingui Qian

- Investigated triboelectric liquid sensing for intelligent prosthetics and developed a distributed sensing scheme based on triboelectric nanogenerators.
- Designed and modeled a tubular liquid sensor in SolidWorks, integrating PTFE and copper electrodes to generate electrical signals during liquid flow.
- Optimized sensor dimensions, performance, and fabrication processes, achieving over 90% sensing accuracy.